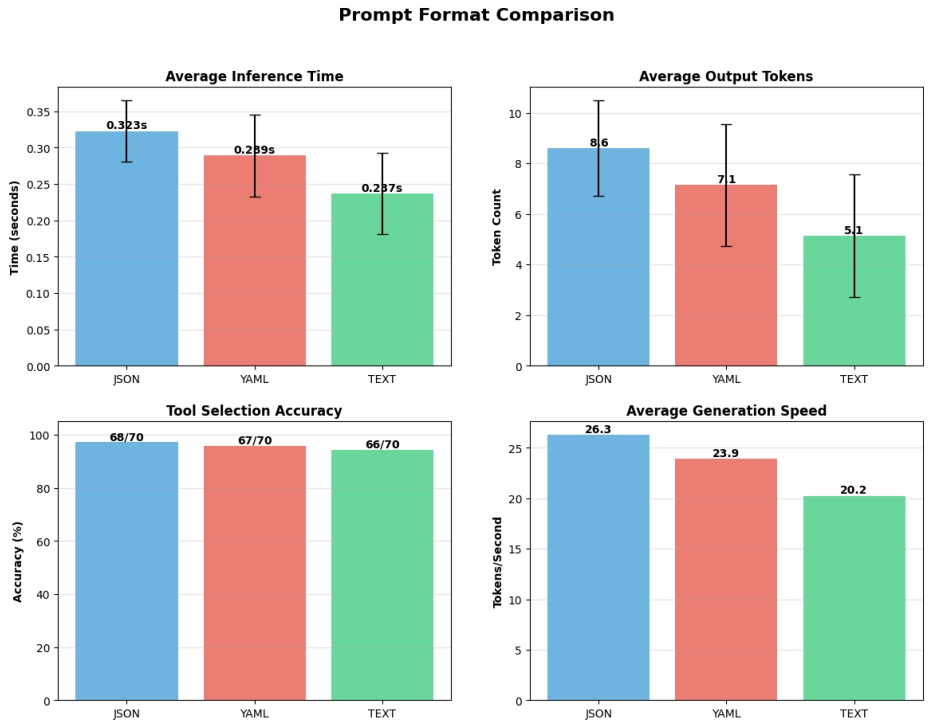
CTE 1.1 VIGIL Agent > TRL 3.6

Vigil Agent Tool Calling Capabilities Test Report



Summary

This report presents the results of comprehensive testing conducted on the VIGIL Agent, a tool router designed for medical upskilling systems. The testing evaluated three different output format configurations (JSON, YAML, and TEXT) across 70 standardized test queries to assess accuracy, performance, and reliability in routing user queries to appropriate tools. These metrics were collected only for the final selected prompt configurations. All tests were run with Qwen2.5-7b-instruct using vLLM.

Key Findings:

- JSON format achieved the highest accuracy at 97.1% (68/70 correct classifications)
- All three formats demonstrated strong performance with >94% accuracy
- TEXT format showed the fastest response time (0.237s average) around 40% faster than the JSON format

1. Test Methodology

1.1 System Under Test

Agent Name: Vigil
Purpose: Tool router for medical ultrasound system
Available Tools:
- `<VQA>` - Visual Question Answering (for ultrasound image interpretation)
- `<GMQA>` - General Medical Question Answering (for medical knowledge queries)
- `[]` - No tool required (for external searches, casual conversation, administrative queries)

1.2 Test Configurations

Three output format variants were tested:
1. JSON Format - Structured JSON response with tools array
2. YAML Format - YAML-formatted response with tools list
3. TEXT Format - Python list format response

1.3 Test Dataset

- Total Test Cases: 70 queries
- Query Distribution:
  - VQA queries: 30 (visual interpretation and scanning technique questions)
  - GMQA queries: 15 (general medical knowledge questions)
  - No-tool queries: 25 (external search requests and casual conversation)

1.4 Evaluation Metrics

- Accuracy: Percentage of correct tool selections
- Response Time: Average time to generate routing decision
- Token Generation: Average tokens produced per response
- Throughput: Tokens generated per second
- Consistency: Standard deviation of timing and token metrics

2. Performance Results

2.1 Summary Statistics

Metric	JSON Format	YAML Format	Text Format
Accuracy	97.1% (68/70)	95.7% (67/70)	94.3% (66/70)
Avg. Response Time	0.323s	0.289s	0.237s
Avg. Tokens Generated	8.6 tokens	7.1 tokens	5.1 tokens

Throughput	26.3 tokens/sec	23.9 tokens/sec	20.2 tokens/sec

2.2 Accuracy Analysis

JSON Format:
- Correct classifications: 68
- Incorrect classifications: 2
- Error rate: 2.9%

YAML Format:
- Correct classifications: 67
- Incorrect classifications: 3
- Error rate: 4.3%

TEXT Format:
- Correct classifications: 66
- Incorrect classifications: 4
- Error rate: 5.7%

2.3 Performance Trade-offs

Speed vs. Accuracy:
- TEXT format was 27% faster than JSON (0.237s vs 0.323s)
- However, JSON format was 3% more accurate than TEXT (97.1% vs 94.3%)
- YAML format represented a middle ground with moderate speed and accuracy

Token Generation Comparison:
- TEXT: 5.1 tokens average (±2.4) - 41% fewer tokens than JSON
- YAML: 7.1 tokens average (±2.4) - 17% fewer tokens than JSON
- JSON: 8.6 tokens average (±1.9) - baseline

Impact on Response Time:
- TEXT format's minimal token output (just a Python list: `["<VQA>"]` or `[]`) eliminated formatting overhead
- JSON's structured format required additional tokens for object syntax: `{"tools": ["<VQA>"]}`
- YAML's syntax (keys and formatting) fell between TEXT and JSON

Cost and Latency Benefits:
1. Reduced Costs: Fewer output tokens = lower per-request costs at scale
2. Faster Time-to-First-Token: Simpler format allows model to begin output sooner
3. Lower Bandwidth: Minimal token output reduces network transfer time

4. Better Throughput: Despite lower tokens/sec (20.2 vs 26.3), TEXT's brevity means faster complete responses

Real-World Impact:
- TEXT format could save ~40% on output token costs compared to JSON
- 0.086s faster average response (27% improvement) enhances user experience

3. Detailed Analysis

3.1 Classification Performance by Query Type

All three formats performed well across different query categories:

VQA Queries (Visual Interpretation)
- Strong performance across all formats
- High accuracy in identifying visual context keywords ("this", "here", "shown")
- Successful routing of both general visual questions and scanning technique queries

GMQA Queries (General Medical Knowledge)
- Excellent discrimination from VQA queries
- Accurate identification of medical knowledge questions without visual components
- Clear understanding of queries requiring training data vs. external information

No-Tool Queries (External/Casual)
- Effective filtering of external search requests
- Proper handling of casual conversation
- Good recognition of temporal indicators ("latest", "2024", "recent")

3.2 Error Analysis

The small number of errors across formats (2-4 per 70 queries) suggests:
- Edge cases where query intent was ambiguous
- Potential overlap between medical knowledge and external information requests
- Possible sensitivity to specific keyword combinations

Common error patterns included:
- Borderline cases between GMQA and no-tool categories
- Queries with subtle temporal implications
- Medical questions that could require current guidelines vs. established knowledge

3.3 Prompt Design Effectiveness

All three prompts shared identical decision logic and critical rules, with only output format differing. The consistently high accuracy (>94%) across formats indicates:

- Robust prompt design with clear category definitions
- Effective use of keywords and decision heuristics
- Well-constructed examples demonstrating edge cases
- Strong adherence to the "select ONLY ONE tool or no tool" constraint

4. Conclusions

The Vigil tool router demonstrates strong performance across all tested configurations, with accuracy consistently exceeding 94%. The testing revealed important trade-offs between format efficiency and accuracy that should inform deployment decisions.

Key Findings:

1. TEXT Format Efficiency: The TEXT format's 41% reduction in output tokens (5.1 vs 8.6 average) and 27% faster response time (0.237s vs 0.323s) demonstrates significant efficiency gains. This format was specifically tested to evaluate whether minimal token output—using simple Python list syntax like `["<VQA>"]` instead of JSON's `{"tools": ["<VQA>"]}`—could deliver meaningful cost and latency improvements without compromising accuracy.

2. Accuracy-Speed Trade-off: While TEXT format achieved impressive efficiency metrics, its 2.8% lower accuracy (94.3% vs 97.1%) translates to 2-4 additional misrouted queries per 70 requests. In medical applications, this accuracy difference is significant enough to warrant careful consideration.

3. Production Recommendation: The JSON format variant is recommended for production deployment due to its superior accuracy (97.1%) and consistent performance characteristics (σ=0.042s), which are essential for medical applications where routing errors could impact clinical workflows. However, for high-volume systems with cost constraints and tolerance for slightly lower accuracy, TEXT format offers compelling efficiency advantages.

4. Prompt Design Validation: The consistently high accuracy (>94%) across all formats validates the effectiveness of the underlying prompt design and decision logic, independent of output format choice.

Appendix A: Test Configuration Details

Prompt Structure
All three formats used identical:
- Decision logic with 3 main routing categories
- Tool descriptions and capabilities
- Critical rules and keyword indicators
- Example query classifications

Test Query Categories

- VQA General Visual (10 queries): "What am I looking at?", "Does anything look abnormal?"
- VQA Scanning Technique (10 queries): "How can I get a better view?", "Is my probe positioned correctly?"
- VQA Specific Findings (10 queries): "Can you measure the size?", "What is the echogenicity?"
- GMQA Medical Knowledge (15 queries): "What are symptoms of diabetes?", "How is anemia classified?"
- No Tool - External Search (15 queries): "Latest guidelines from 2024", "Search for recent studies"
- No Tool - Casual (10 queries): "Hello", "Thanks", "Tell me a joke"

Testing Environment
- Consistent model and temperature settings across all formats
- Identical test query sequence and ordering
- Controlled timing measurements for fair comparison